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lab4.py X
C: > aiml > lab4.py
1 class RuleBasedSystem:
2     def __init__(self, facts, rules):
3         self.facts = set(facts)
4         self.rules = rules
5
6     def forward_chain(self):
7         iterations = 0
8
9         while True:
10             new_fact_added = False
11
12             for rule in self.rules:
13                 # Check if all conditions of the rule exist in current facts
14                 if all(cond in self.facts for cond in rule['if']):
15                     if rule['then'] not in self.facts:
16                         print(
17                             f"Rule Triggered: IF {rule['if']} "
18                             f"THEN Add {rule['then']}"
19                         )
20
21                     self.facts.add(rule['then'])
22                     new_fact_added = True
23
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\madan\AppData\Local\Programs\Microsoft VS Code> & C:\Python314\python.exe c:/aiml/lab4.py
Initial Facts: ['Socrates_is_human', 'All_humans_are_mortal']
Rule Triggered: IF ['Socrates_is_human', 'All_humans_are_mortal'] THEN Add Socrates_is_mortal
Final Knowledge Base Facts: {'Socrates_is_mortal', 'All_humans_are_mortal', 'Socrates_is_human'}
PS C:\Users\madan\AppData\Local\Programs\Microsoft VS Code> 
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OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\madan\AppData\Local\Programs\Microsoft VS Code> & C:\Python314\python.exe c:/aiml/lab4.py
Initial Facts: ['Socrates is human', 'All humans are mortal']
Rule Triggered: IF ['Socrates is human', 'All humans are mortal'] THEN Add Socrates is mortal
Final Knowledge Base Facts: {'Socrates is mortal', 'All humans are mortal', 'Socrates is human'}
PS C:\Users\madan\AppData\Local\Programs\Microsoft VS Code> & C:\Python314\python.exe c:/aiml/lab4.py
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RULE BASED PLANNING SYSTEM
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Initial State: {'has_money': True, 'has_car': False, 'at_home': True, 'at_work': False}
Goal State: {'at_work': True}

Rule Triggered: IF has_money=True AND has_car=False THEN ADD has_car=True
Current State: {'has_money': False, 'has_car': True, 'at_home': True, 'at_work': False}

Rule Triggered: IF has_car=True AND at_home=True THEN ADD at_work=True
Current State: {'has_money': False, 'has_car': True, 'at_home': False, 'at_work': True}

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Goal Reached!

Execution Plan:
1. Buy Car
2. Drive to Work
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PS C:\Users\madan\AppData\Local\Programs\Microsoft VS Code> & C:\Python314\python.exe c:/aiml/lab4.py
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RULE BASED PLANNING SYSTEM
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Initial State: {'has_money': True, 'has_car': False, 'at_home': True, 'at_work': False}
Goal State: {'at_work': True}

Rule Triggered: IF has_money=True AND has_car=False THEN ADD has_car=True
Current State: {'has_money': False, 'has_car': True, 'at_home': True, 'at_work': False}

Rule Triggered: IF has_car=True AND at_home=True THEN ADD at_work=True
Current State: {'has_money': False, 'has_car': True, 'at_home': False, 'at_work': True}

=====
Goal Reached!

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Lab5

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aiml > lab5.py > find_s
1 def find_s(training_data):
2     # Initialize the specific hypothesis with the first positive instance
3     hypothesis = None
4     for row in training_data:
5         if row[-1] == 'Yes': # Target concept is positive
6             hypothesis = row[:-1].copy()
7             break
8
9     if hypothesis is None:
10        return "No positive instances found."
11
12    # Update hypothesis based on remaining positive instances
13    for row in training_data:
14        if row[-1] == 'Yes':
15            for i in range(len(hypothesis)):
16                if row[i] != hypothesis[i]:
17                    hypothesis[i] = '?'
18
19    return hypothesis
20
21 # Example Usage
22 if __name__ == "__main__":
23     # Columns: Sky, Temp, Humidity, Wind, Water, Forecast, EnjoySport
24     dataset = [
25         ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
26         ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
27         ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
28         ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
29     ]
30
31     most_specific_hypothesis = find_s(dataset)
32     print("Most Specific Hypothesis found by Find-S:", most_specific_hypothesis)

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PS C:\Users\madan\AppData\Local\Programs\Microsoft VS Code> & C:\Python314\python.exe c:/aiml/lab4.py
Initial Facts: ['Socrates_is_human', 'All_humans_are_mortal']
Rule Triggered: IF ['Socrates_is_human', 'All_humans_are_mortal'] THEN Add Socrates_is_mortal
Final Knowledge Base Facts: {'Socrates_is_mortal', 'All_humans_are_mortal', 'Socrates_is_human'}
PS C:\Users\madan\AppData\Local\Programs\Microsoft VS Code> & C:\Python314\python.exe c:/aiml/lab4.py
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RULE BASED PLANNING SYSTEM

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Initial State: {'has_money': True, 'has_car': False, 'at_home': True, 'at_work': False}
Goal State: {'at_work': True}

Rule Triggered: IF has_money=True AND has_car=False THEN ADD has_car=True
Current State: {'has_money': False, 'has_car': True, 'at_home': True, 'at_work': False}

Rule Triggered: IF has_car=True AND at_home=True THEN ADD at_work=True
Current State: {'has_money': False, 'has_car': True, 'at_home': False, 'at_work': True}

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Goal Reached!

Execution Plan:
1. Buy Car
2. Drive to Work

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PS C:\Users\madan\AppData\Local\Programs\Microsoft VS Code> & C:\Python314\python.exe c:/aiml/lab4.py
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RULE BASED PLANNING SYSTEM

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Initial State: {'has_money': True, 'has_car': False, 'at_home': True, 'at_work': False}
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Goal Reached!

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Goal Reached!

Execution Plan:
1. Buy Car
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PS C:\Users\madan\AppData\Local\Programs\Microsoft VS Code> & C:\Python314\python.exe c:/aiml/lab5.py
Most Specific Hypothesis found by Find-S: ['Sunny', 'Warm', '?', 'Strong', '?', '?']
PS C:\Users\madan\AppData\Local\Programs\Microsoft VS Code> █
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